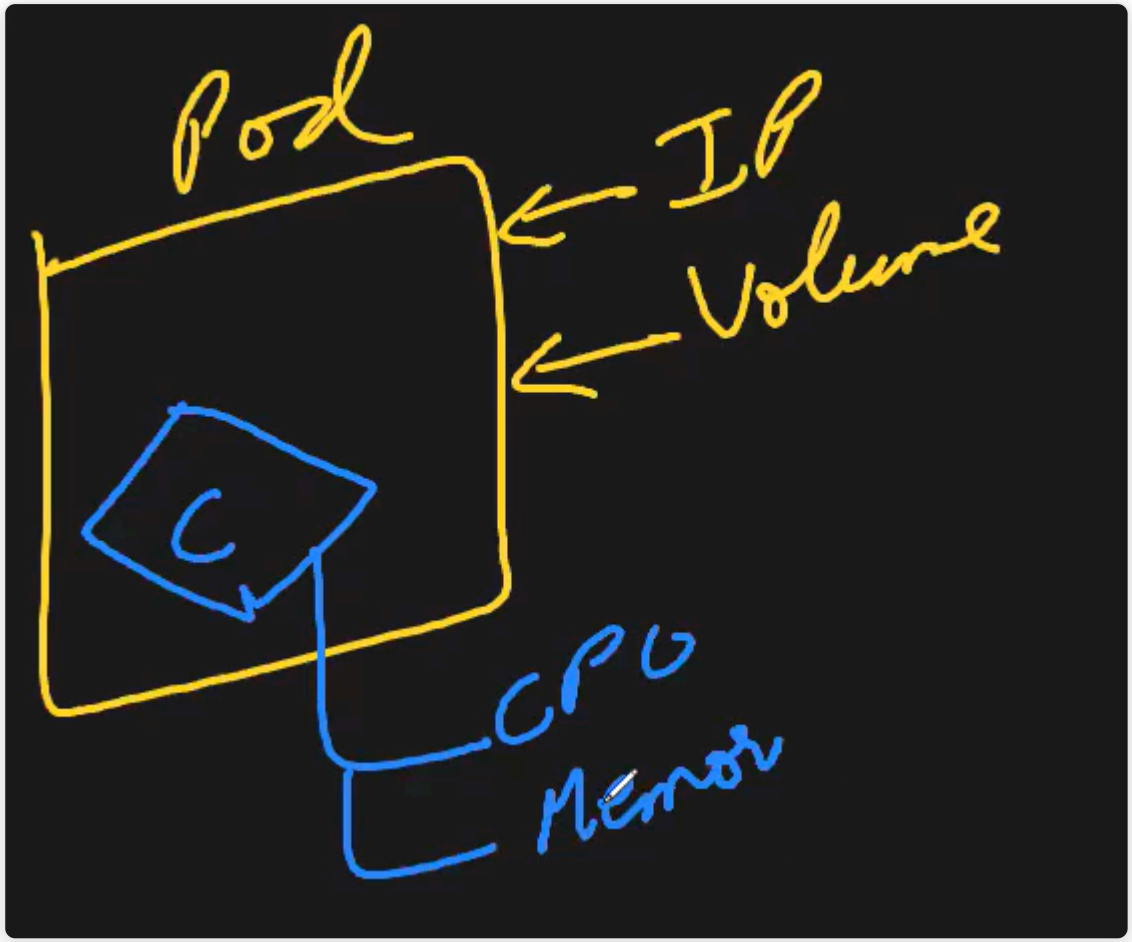
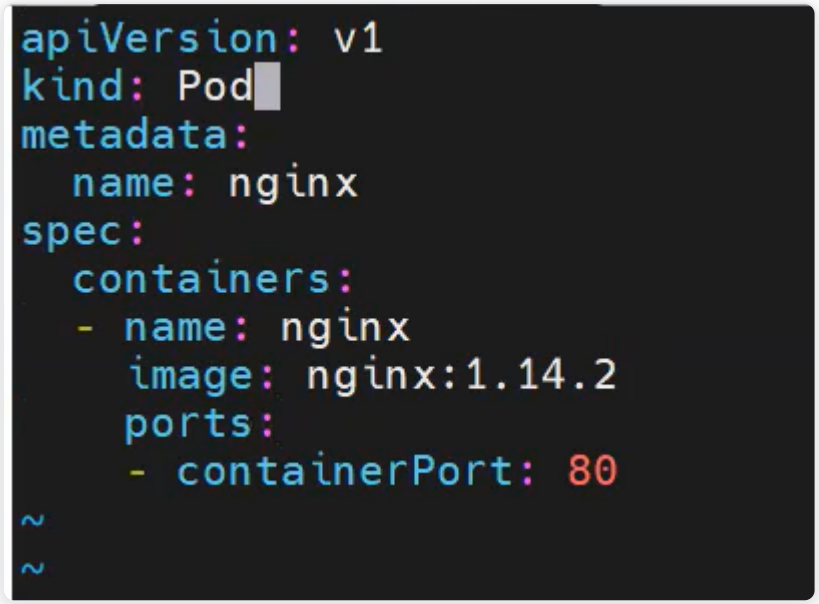
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 01

Screenshots: Importance of Labels & Running Pods

Raw screenshots captured while watching this session, kept alongside the written notes (01-labels-and-running-pods.md) for visual reference during interview revision.

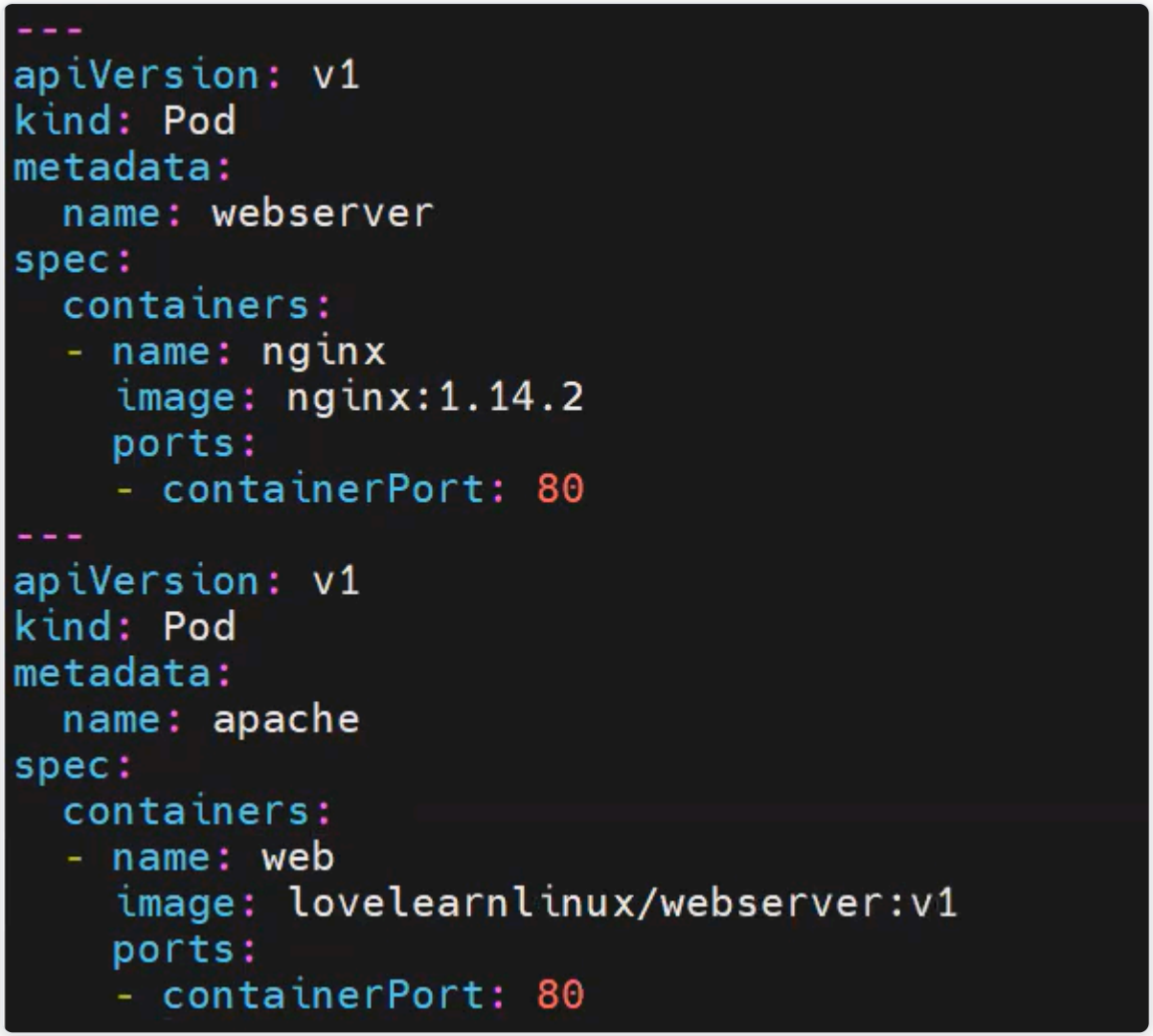
33 screenshots · captured 2026-09-10, 12:33 PM - 1:51 PM


Pod
IP
Volume
C
CPU
Memor


apiVersion: v1
kind: Pod
metadata:
name: nginx
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
~
~

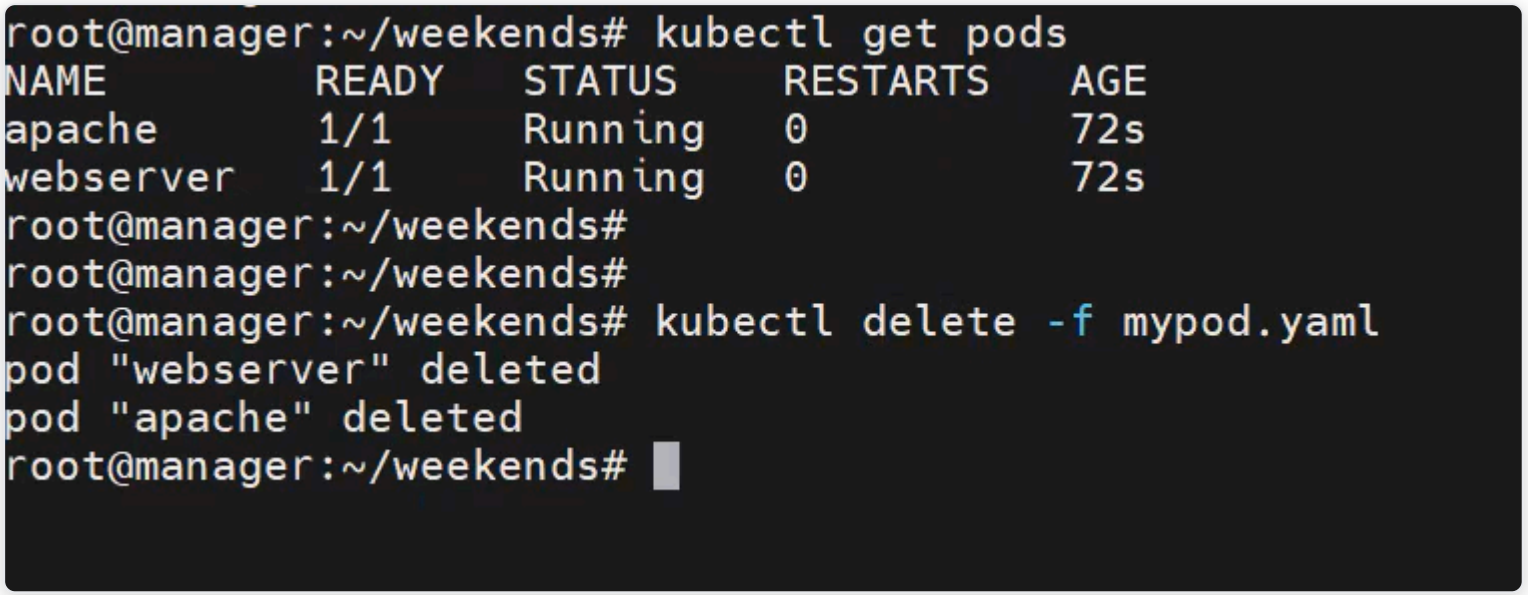
```
kubectl create -f mypod.yaml
```

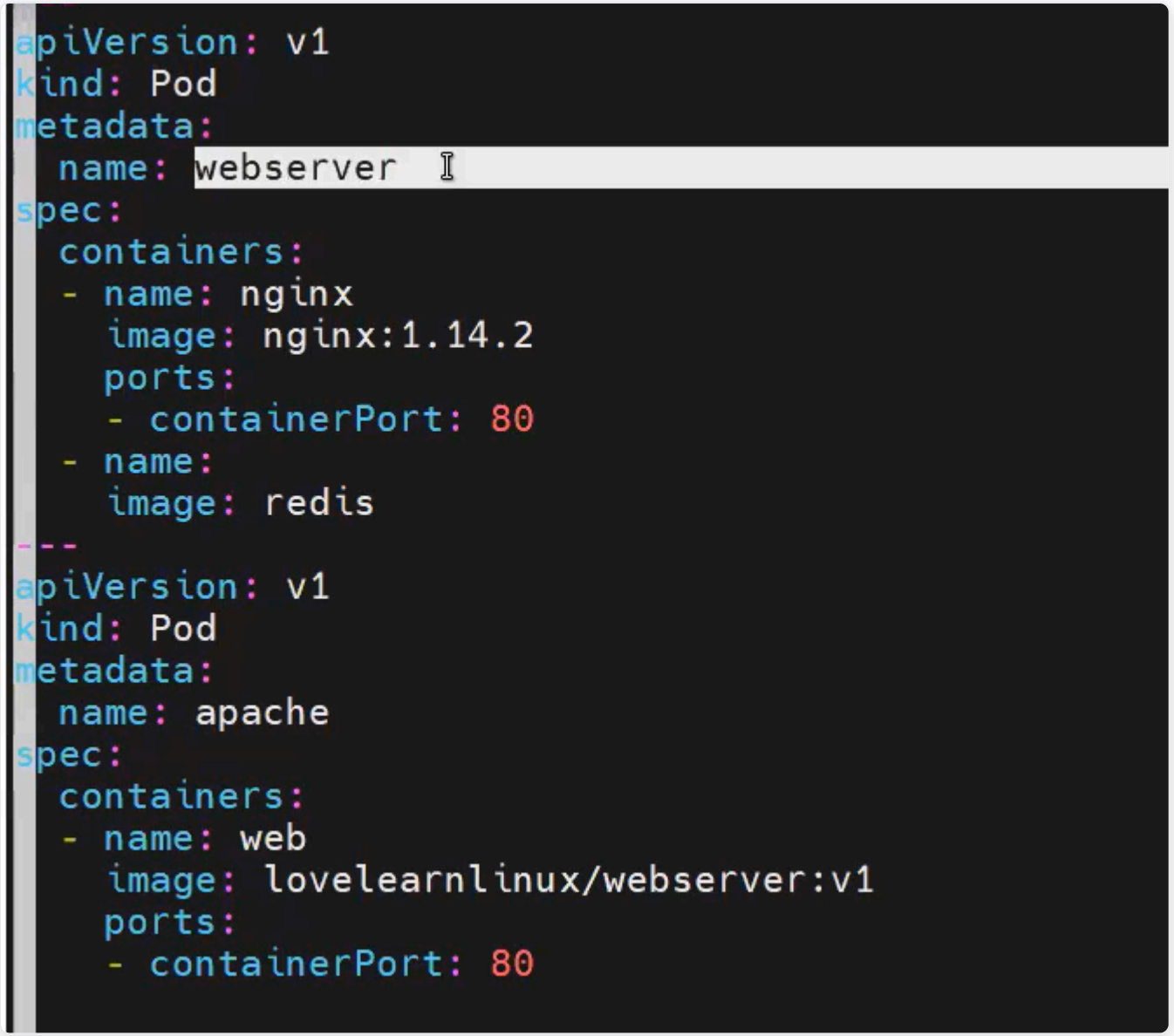
```
root@manager:~/weekends# kubectl create -f mypod.yaml
pod/webserver created
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
webserver   1/1     Running   0          7s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get nodes
NAME      STATUS   ROLES           AGE     VERSION
manager   Ready    control-plane   3d22h   v1.27.1
nodeone   Ready    <none>          3d22h   v1.27.1
nodetwo   Ready    <none>          29h     v1.26.1
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
webserver   1/1     Running   0          57s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP              NODE      NOMINATED NODE   READINESS GATES
webserver   1/1     Running   0          63s   172.16.3.129    nodetwo   <none>           <none>
root@manager:~/weekends# kubectl describe pod webserver
```

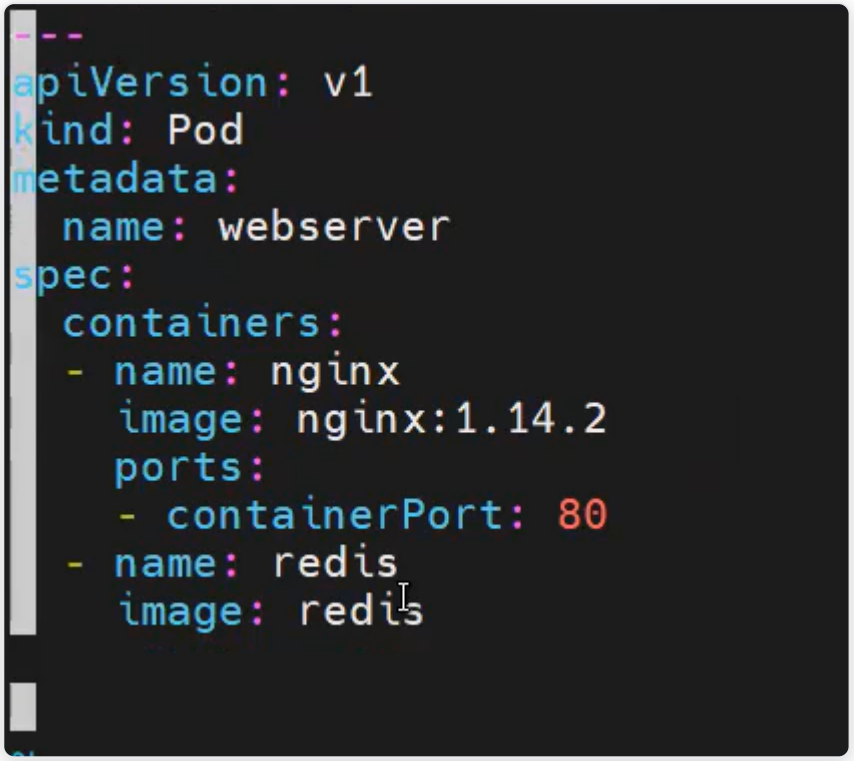

---
apiVersion: v1
kind: Pod
metadata:
name: webserver
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
name: apache
spec:
containers:
- name: web
image: lovelearnlinux/webserver:v1
ports:
- containerPort: 80

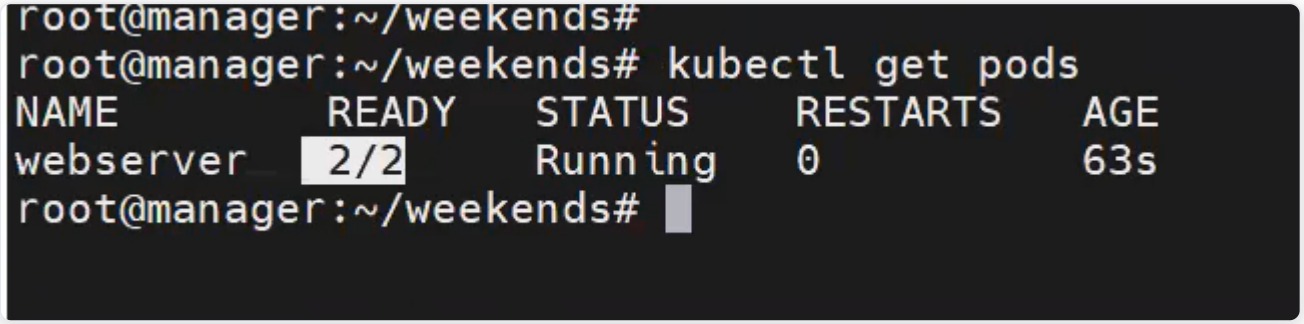
```
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f mypod.yaml
pod/webserver created
pod/apache created
root@manager:~/weekends#
root@manager:~/weekends# kubect
```

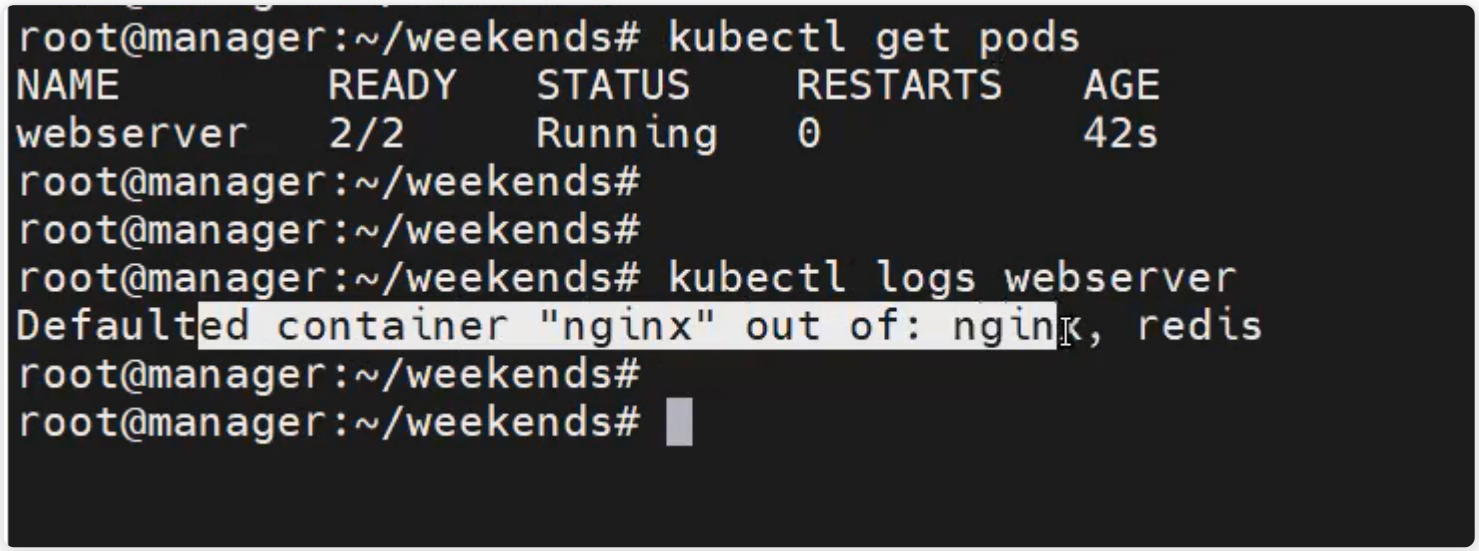
```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f mypod.yaml
pod/webserver created
pod/apache created
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
apache      1/1     Running   0          6s
webserver   1/1     Running   0          6s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods  -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP              NODE      NOMINATED NODE   READINESS GATES
apache      1/1     Running   0          9s    172.16.3.132    nodetwo   <none>           <none>
webserver   1/1     Running   0          9s    172.16.3.131    nodetwo   <none>           <none>
root@manager:~/weekends#
root@manager:~/weekends#
```

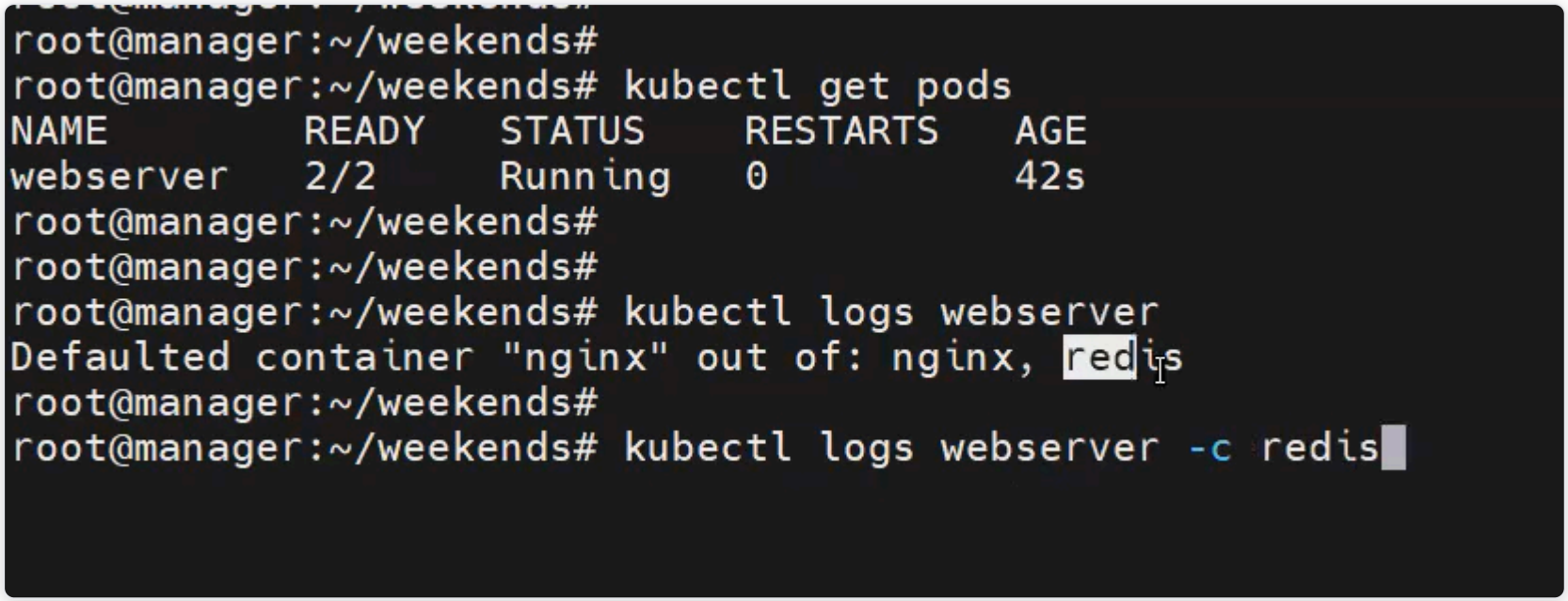

root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
apache 1/1 Running 0 72s
webserver 1/1 Running 0 72s
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl delete -f mypod.yaml
pod "webserver" deleted
pod "apache" deleted
root@manager:~/weekends#

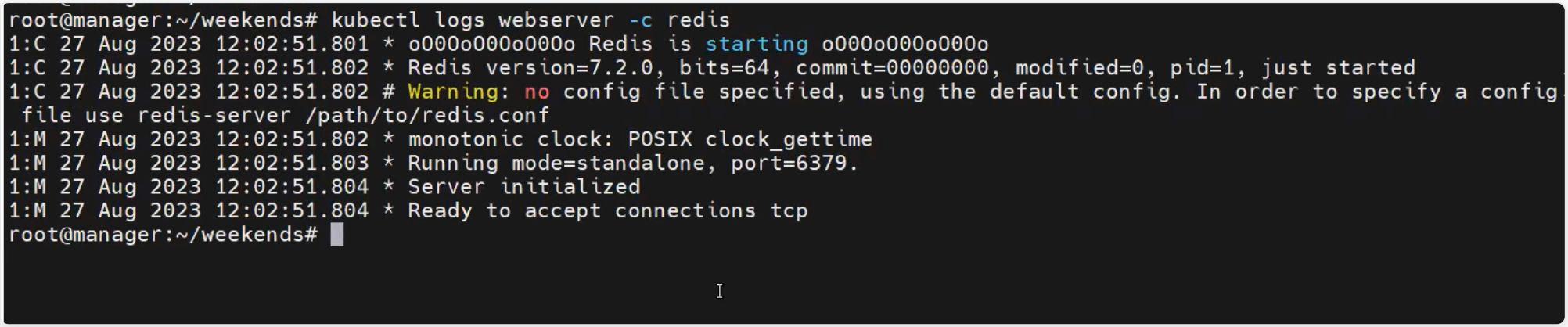

apiVersion: v1
kind: Pod
metadata:
name: webserver
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
- name:
image: redis
---
apiVersion: v1
kind: Pod
metadata:
name: apache
spec:
containers:
- name: web
image: lovelearnlinux/webserver:v1
ports:
- containerPort: 80

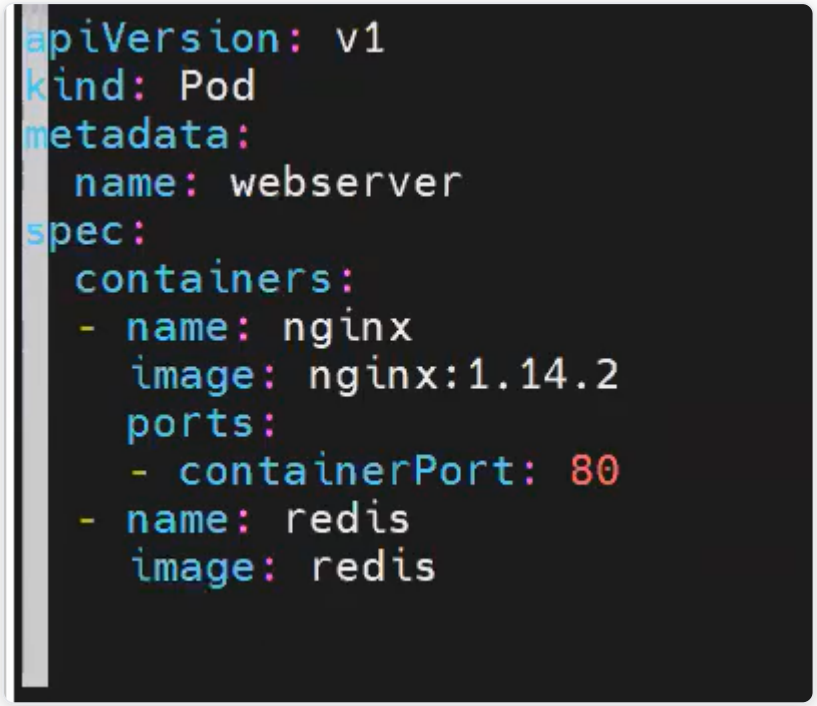

---
apiVersion: v1
kind: Pod
metadata:
name: webserver
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
- name: redis
image: redis

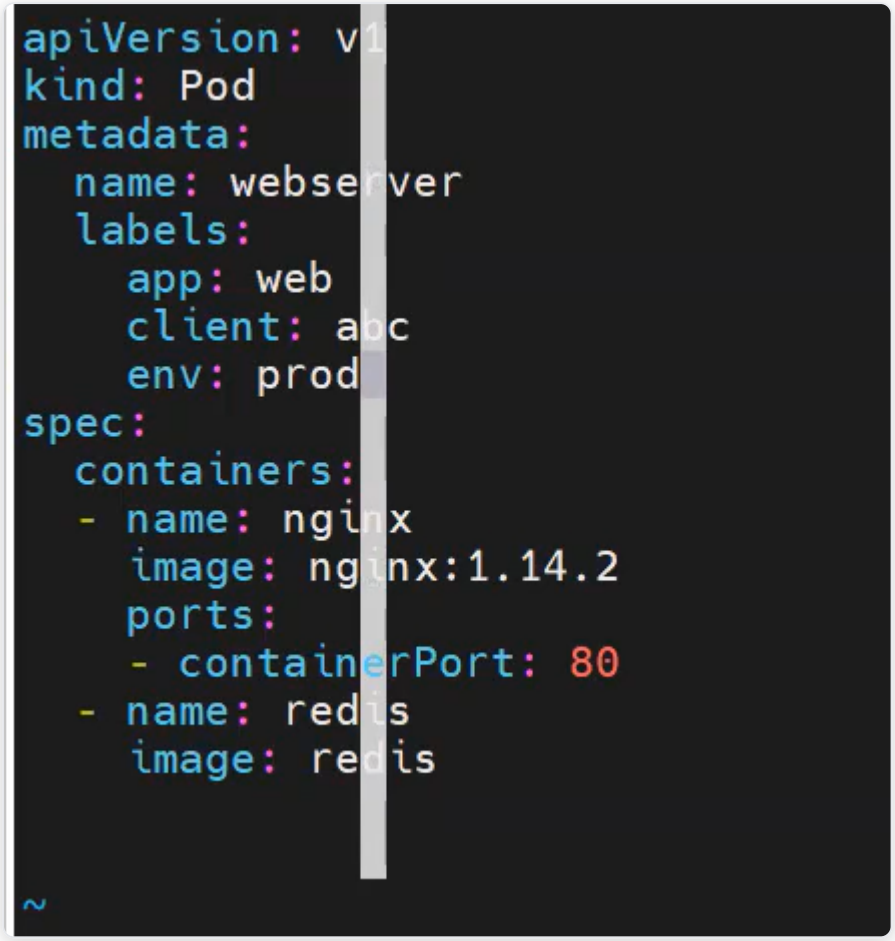

root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
webserver 2/2 Running 0 63s
root@manager:~/weekends#

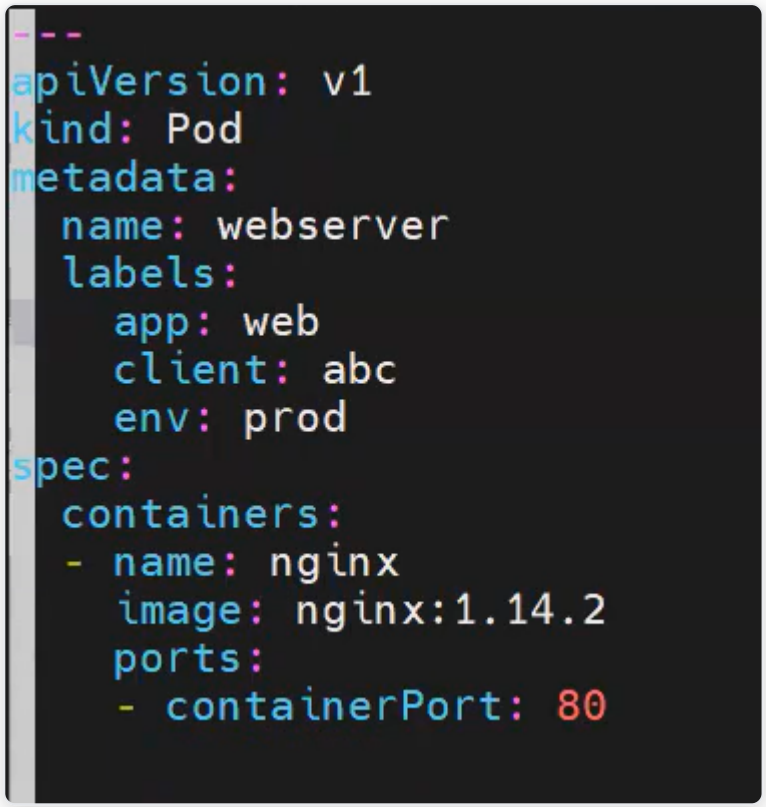

root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
webserver 2/2 Running 0 42s
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl logs webserver
Defaulted container "nginx" out of: nginx, redis
root@manager:~/weekends#
root@manager:~/weekends#

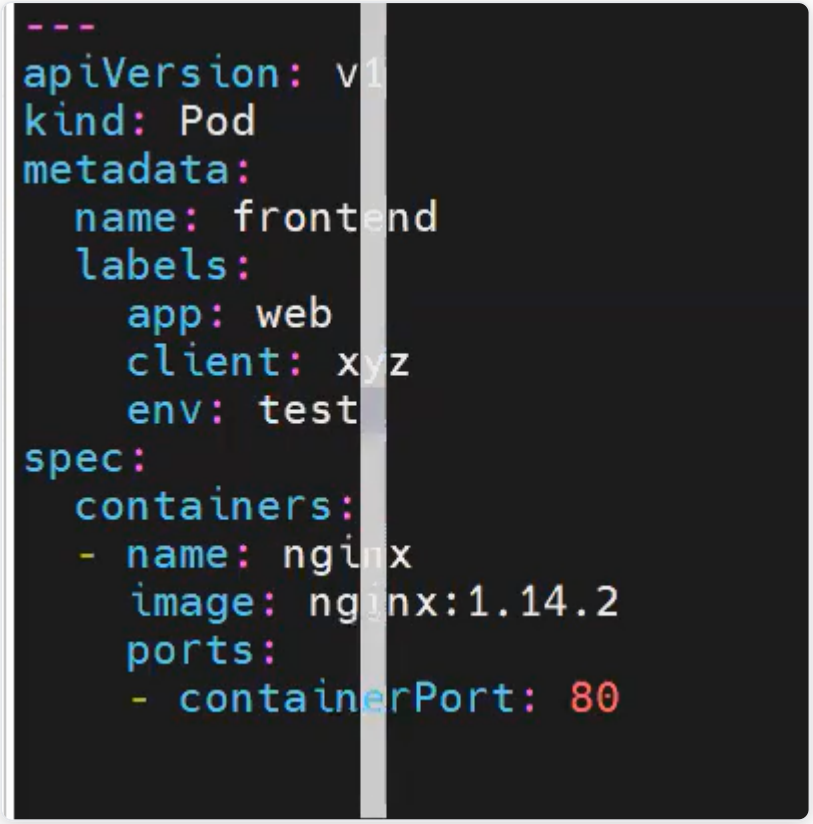

root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
webserver 2/2 Running 0 42s
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl logs webserver
Defaulted container "nginx" out of: nginx, redis
root@manager:~/weekends#
root@manager:~/weekends# kubectl logs webserver -c redis

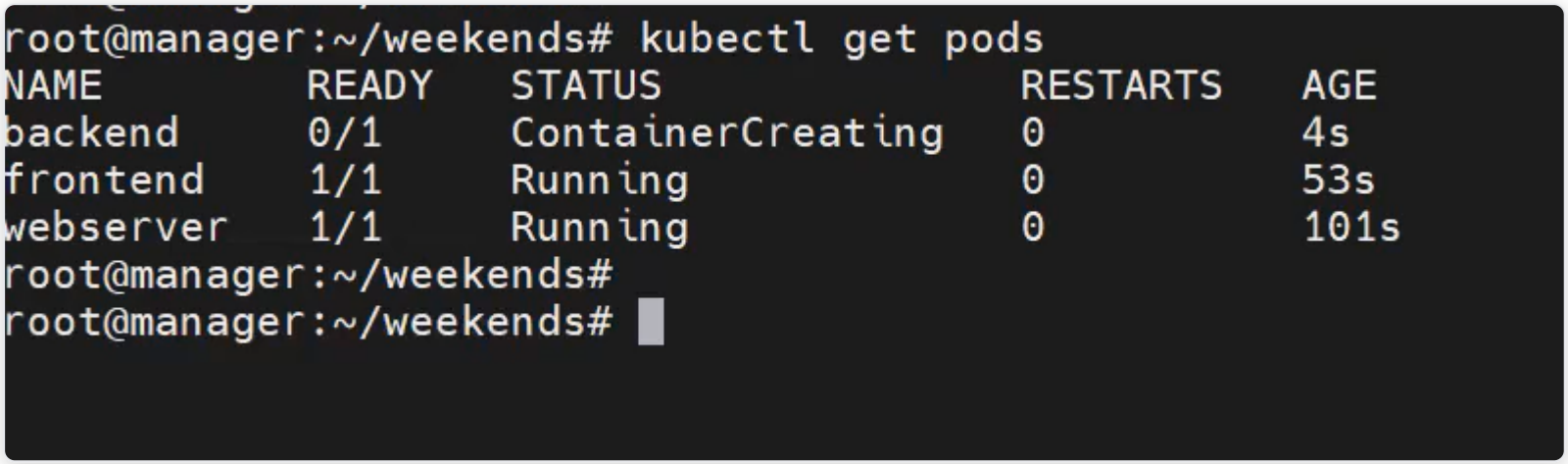

root@manager:~/weekends# kubectl logs webserver -c redis
1:C 27 Aug 2023 12:02:51.801 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 27 Aug 2023 12:02:51.802 * Redis version=7.2.0, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 27 Aug 2023 12:02:51.802 # Warning: no config file specified, using the default config. In order to specify a config
file use redis-server /path/to/redis.conf
1:M 27 Aug 2023 12:02:51.802 * monotonic clock: POSIX clock_gettime
1:M 27 Aug 2023 12:02:51.803 * Running mode=standalone, port=6379.
1:M 27 Aug 2023 12:02:51.804 * Server initialized
1:M 27 Aug 2023 12:02:51.804 * Ready to accept connections tcp
root@manager:~/weekends#

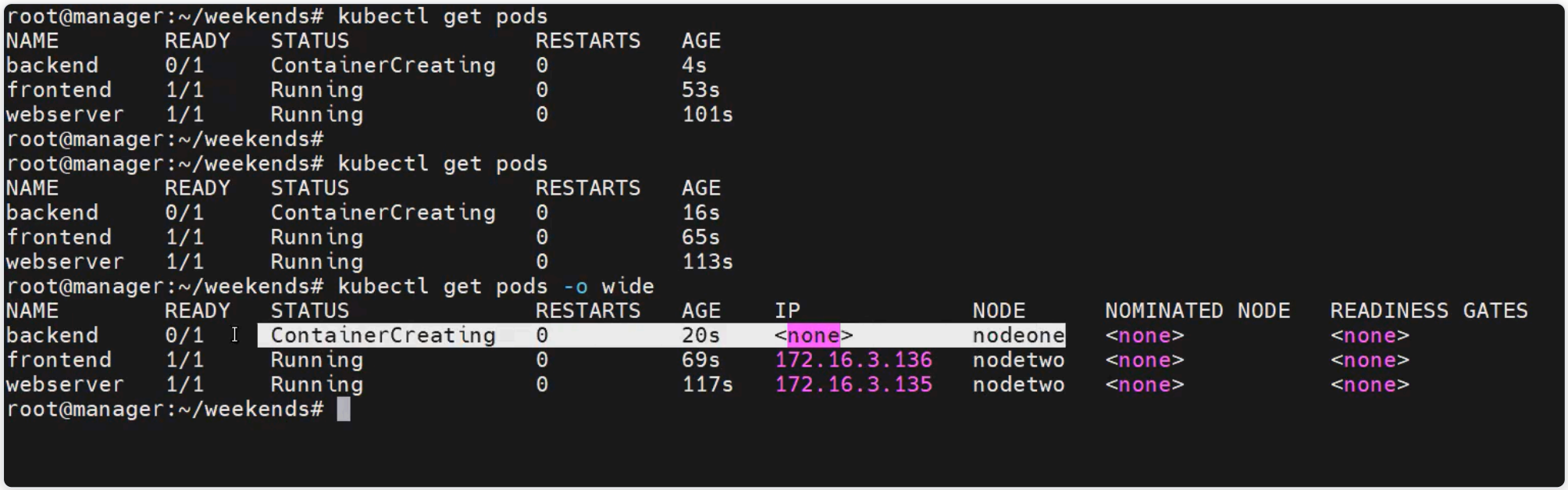

apiVersion: v1
kind: Pod
metadata:
name: webserver
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
- name: redis
image: redis

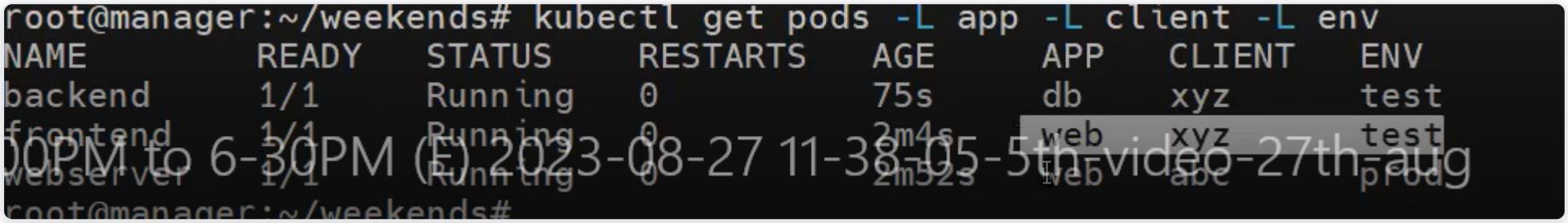

apiVersion: v1
kind: Pod
metadata:
name: webserver
labels:
app: web
client: abc
env: prod
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
- name: redis
image: redis
~


---
apiVersion: v1
kind: Pod
metadata:
name: webserver
labels:
app: web
client: abc
env: prod
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80


---
apiVersion: v1
kind: Pod
metadata:
name: frontend
labels:
app: web
client: xyz
env: test
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80


root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
backend 0/1 ContainerCreating 0 4s
frontend 1/1 Running 0 53s
webserver 1/1 Running 0 101s
root@manager:~/weekends#
root@manager:~/weekends#

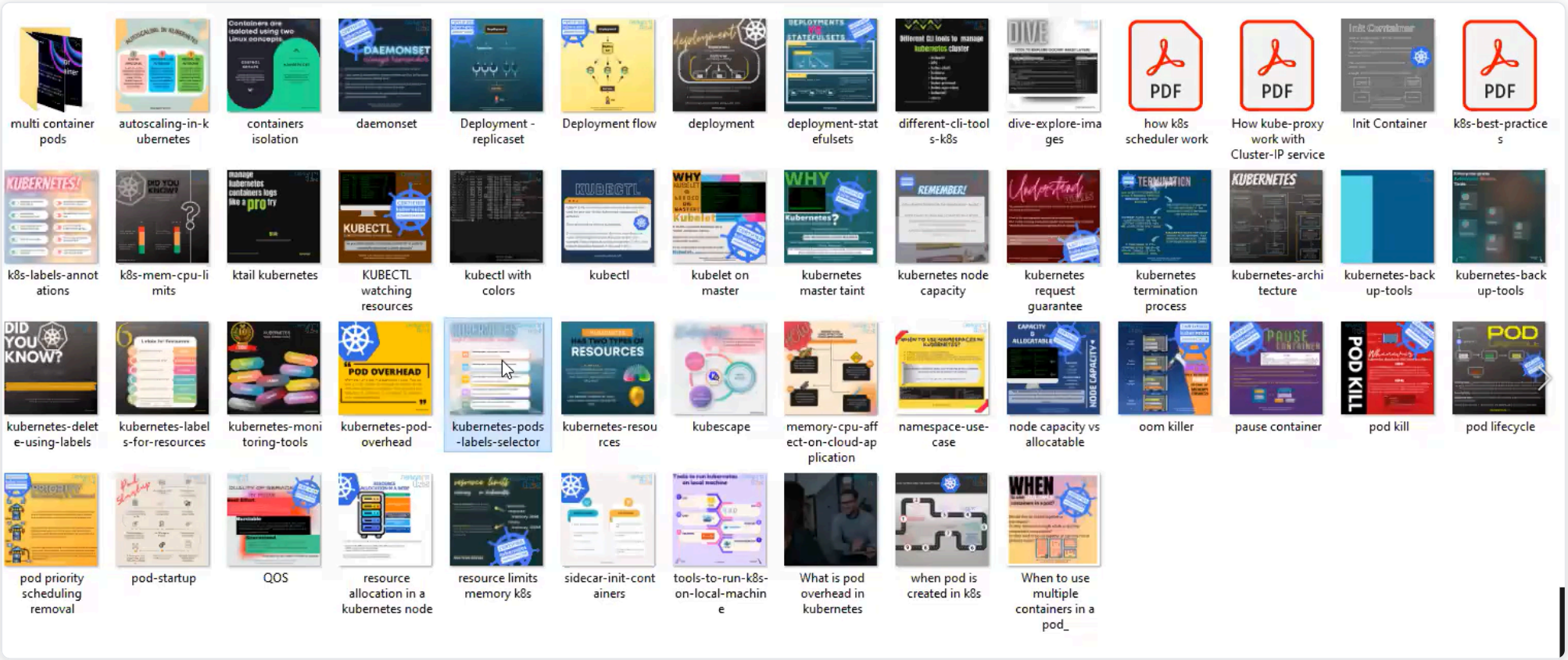

root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
backend 0/1 ContainerCreating 0 4s
frontend 1/1 Running 0 53s
webserver 1/1 Running 0 101s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods
NAME READY STATUS RESTARTS AGE
backend 0/1 ContainerCreating 0 16s
frontend 1/1 Running 0 65s
webserver 1/1 Running 0 113s
root@manager:~/weekends# kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
backend 0/1 ContainerCreating 0 20s <none> nodeone <none> <none>
frontend 1/1 Running 0 69s 172.16.3.136 nodetwo <none> <none>
webserver 1/1 Running 0 117s 172.16.3.135 nodetwo <none> <none>
root@manager:~/weekends#


root@manager:~/weekends# kubectl get pods -L app -L client -L env
NAME READY STATUS RESTARTS AGE APP CLIENT ENV
backend 1/1 Running 0 75s db xyz test
frontend 1/1 Running 0 2m4s web xyz test
webserver 1/1 Running 0 2m52s web abc prod
root@manager:~/weekends#
00PM to 6-30PM (E) 2023-08-27 11-38-05-5th-video-27th-aug

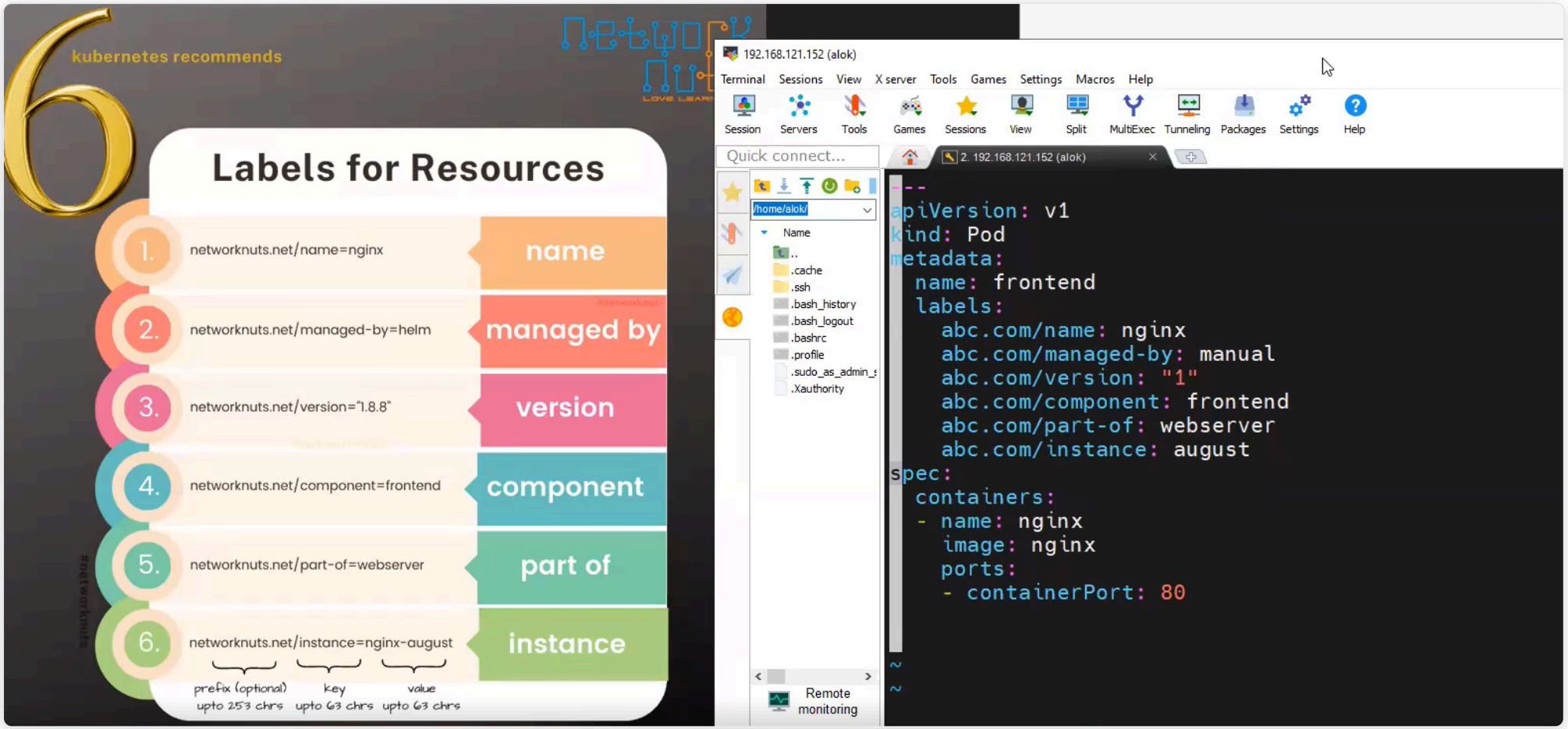

root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -L app -L client -L env
NAME READY STATUS RESTARTS AGE APP CLIENT ENV
backend 1/1 Running 0 85s db xyz test
frontend 1/1 Running 0 2m14s web xyz test
webserver 1/1 Running 0 3m2s web abc prod
root@manager:~/weekends#

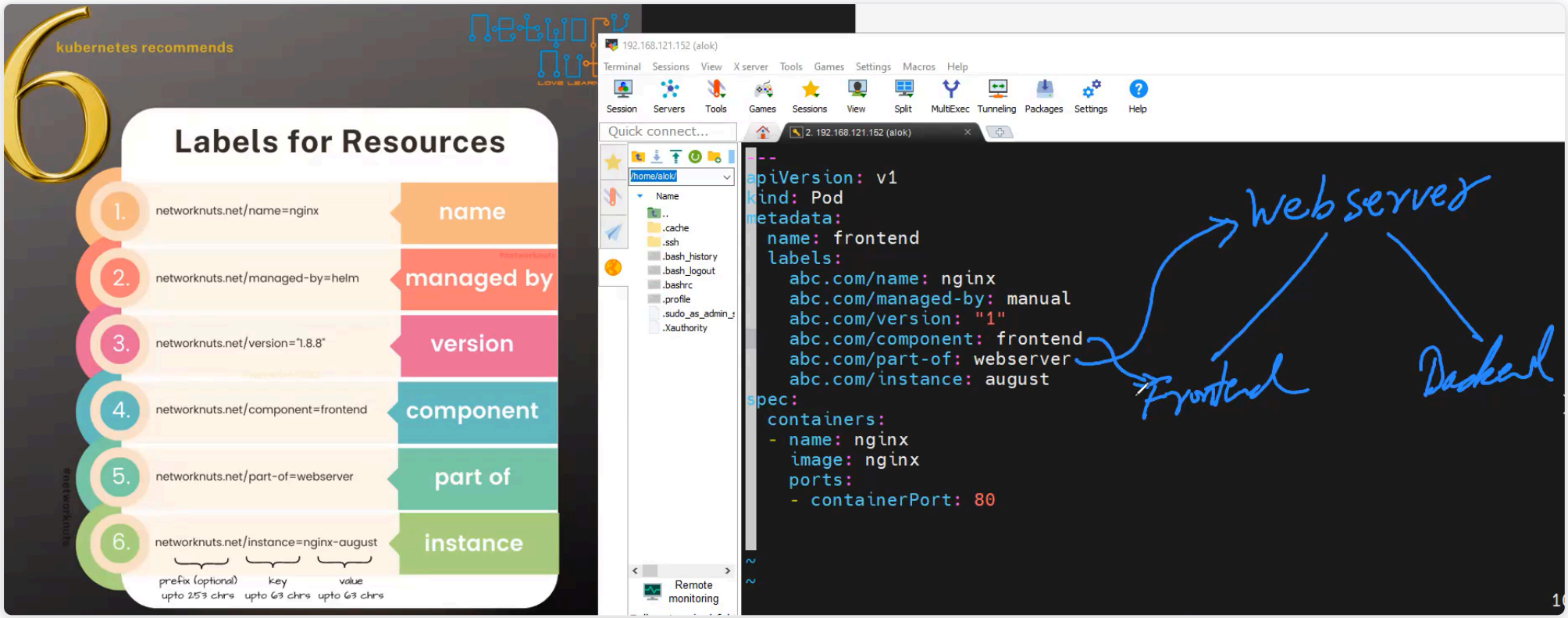
```
root@manager:~/weekends# kubectl get pods -L app -L client -L env
NAME        READY   STATUS    RESTARTS   AGE     APP   CLIENT   ENV
backend     1/1     Running   0          85s     db    xyz      test
frontend    1/1     Running   0          2m14s   web   xyz      test
webserver   1/1     Running   0          3m2s    web   abc      prod
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods --show-labels
NAME        READY   STATUS    RESTARTS   AGE     LABELS
backend     1/1     Running   0          101s    app=db,client=xyz,env=test
frontend    1/1     Running   0          2m30s   app=web,client=xyz,env=test
webserver   1/1     Running   0          3m18s   app=web,client=abc,env=prod
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -l "client=xyz"
NAME       READY   STATUS    RESTARTS   AGE
backend    1/1     Running   0          2m5s
frontend   1/1     Running   0          2m54s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -l "client=xyz" --show-labels
NAME       READY   STATUS    RESTARTS   AGE     LABELS
backend    1/1     Running   0          2m12s   app=db,client=xyz,env=test
frontend   1/1     Running   0          3m1s    app=web,client=xyz,env=test
root@manager:~/weekends#
```

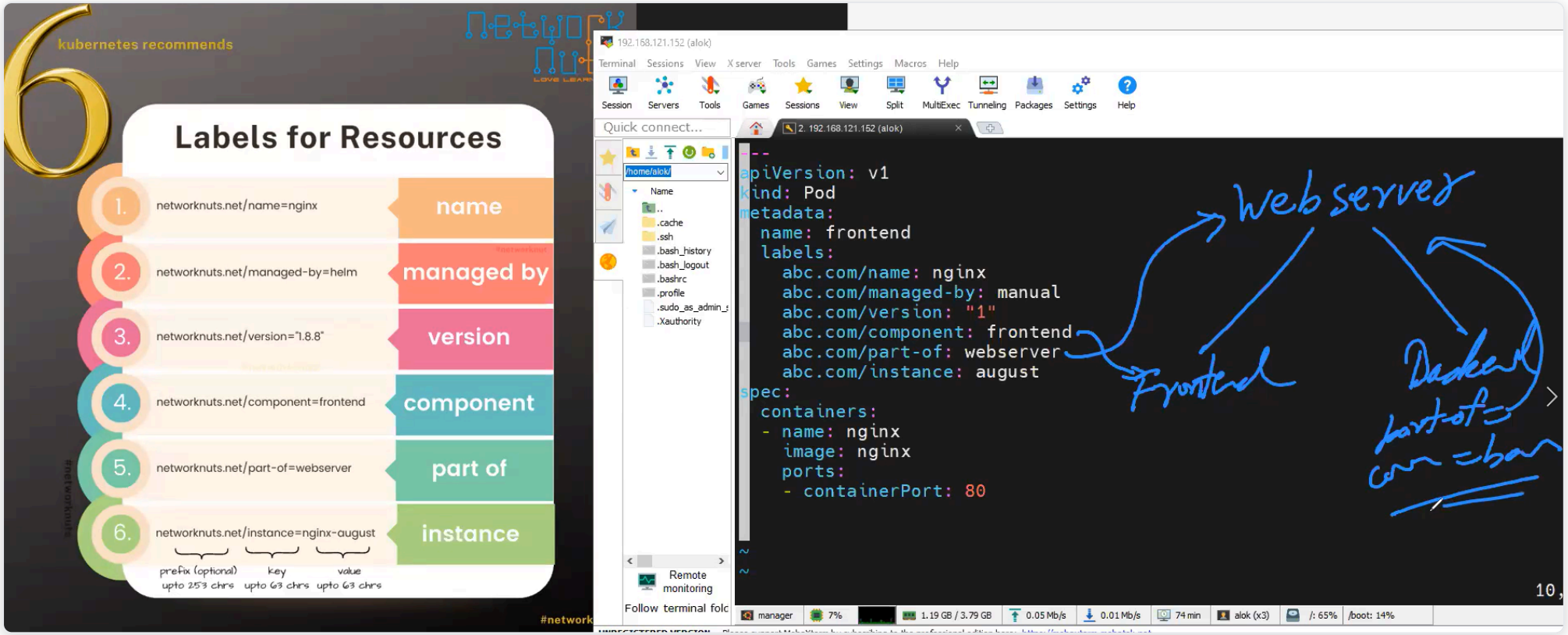
```
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods --show-labels
NAME        READY   STATUS    RESTARTS   AGE     LABELS
backend     1/1     Running   0          3m3s    app=db,client=xyz,env=test
frontend    1/1     Running   0          3m52s   app=web,client=xyz,env=test
webserver   1/1     Running   0          4m40s   app=web,client=abc,env=prod
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl delete pod -l "client=xyz"
pod "backend" deleted
pod "frontend" deleted
root@manager:~/weekends#
```

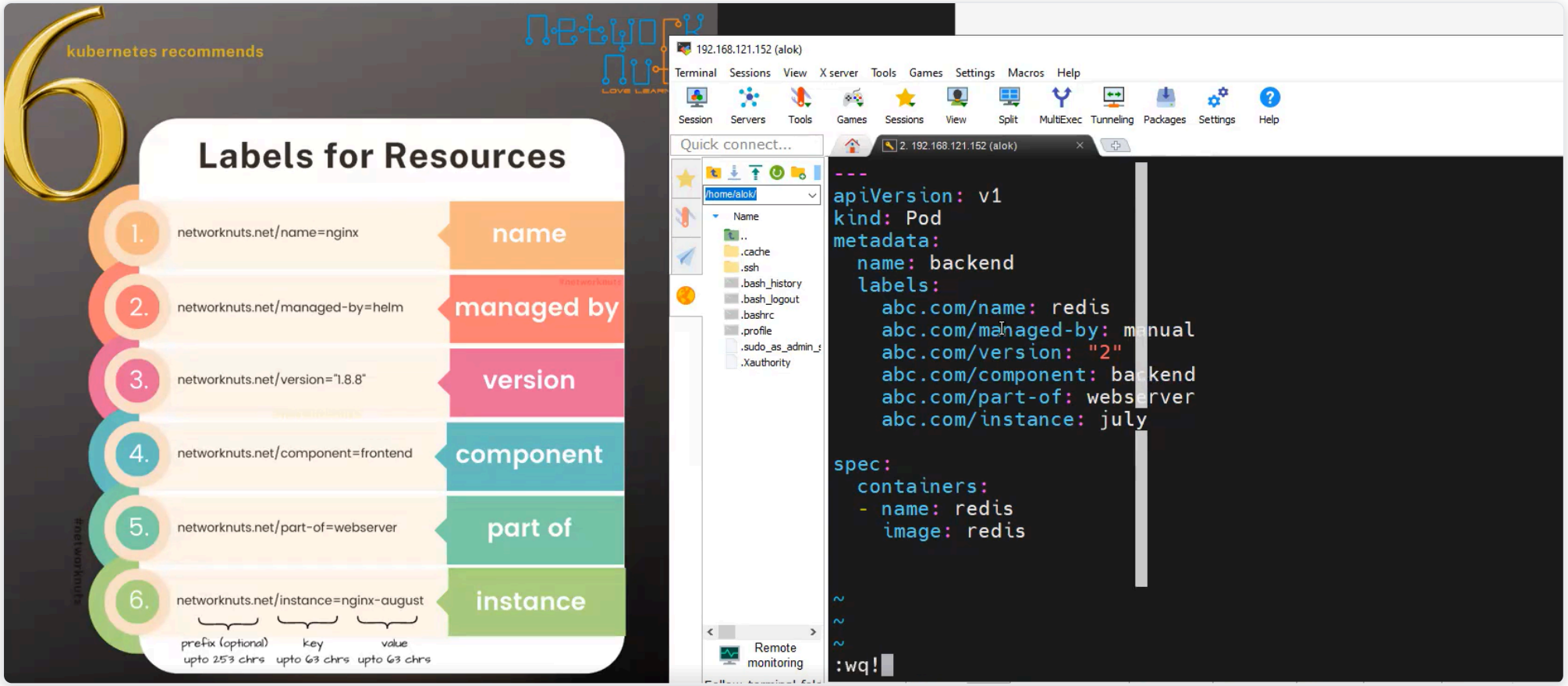

multi container pods
autoscaling-in-kubernetes
containers isolation
daemonset
Deployment - replicaset
Deployment flow
deployment
deployment-statefulsets
different-cli-tools-k8s
dive-explore-images
how k8s scheduler work
How kube-proxy work with Cluster-IP service
Init Container
k8s-best-practices
k8s-labels-annotations
k8s-mem-cpu-limits
ktail kubernetes
KUBECTL watching resources
kubectl with colors
kubectl
kubelet on master
kubernetes master taint
kubernetes node capacity
kubernetes request guarantee
kubernetes termination process
kubernetes-architecture
kubernetes-backup-tools
kubernetes-backup-tools
kubernetes-delete-using-labels
kubernetes-labels-for-resources
kubernetes-monitoring-tools
kubernetes-pod-overhead
kubernetes-pods-labels-selector
kubernetes-resources
kubescape
memory-cpu-affect-on-cloud-application
namespace-use-case
node capacity vs allocatable
oom killer
pause container
pod kill
pod lifecycle
pod priority scheduling removal
pod-startup
QOS
resource allocation in a kubernetes node
resource limits memory k8s
sidecar-init-containers
tools-to-run-k8s-on-local-machine
What is pod overhead in kubernetes
when pod is created in k8s
When to use multiple containers in a pod_


6
kubernetes recommends
Labels for Resources
1.
networknuts.net/name=nginx
name
2.
networknuts.net/managed-by=helm
managed by
3.
networknuts.net/version="1.8.8"
version
4.
networknuts.net/component=frontend
component
5.
networknuts.net/part-of=webserver
part of
6.
networknuts.net/instance=nginx-august
instance
prefix (optional)
upto 253 chrs
key
upto 63 chrs
value
upto 63 chrs
#networknuts


6
kubernetes recommends
Labels for Resources
1. networknuts.net/name=nginx
name
2. networknuts.net/managed-by=helm
managed by
3. networknuts.net/version="1.8.8"
version
4. networknuts.net/component=frontend
component
5. networknuts.net/part-of=webserver
part of
6. networknuts.net/instance=nginx-august
instance
prefix (optional) upto 253 chrs
key upto 63 chrs
value upto 63 chrs
#networknuts
192.168.121.152 (alok)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
2. 192.168.121.152 (alok)
/home/alok/
Name
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.Xauthority
Remote monitoring
---
apiVersion: v1
kind: Pod
metadata:
name: frontend
labels:
abc.com/name: nginx
abc.com/managed-by: manual
abc.com/version: "1"
abc.com/component: frontend
abc.com/part-of: webserver
abc.com/instance: august
spec:
containers:
- name: nginx
image: nginx
ports:
- containerPort: 80


kubernetes recommends
6
Labels for Resources
1. networknuts.net/name=nginx
name
2. networknuts.net/managed-by=helm
managed by
3. networknuts.net/version="1.8.8"
version
4. networknuts.net/component=frontend
component
5. networknuts.net/part-of=webserver
part of
6. networknuts.net/instance=nginx-august
instance
prefix (optional) upto 253 chrs
key upto 63 chrs
value upto 63 chrs
192.168.121.152 (alok)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
2. 192.168.121.152 (alok)
/home/alok/
Name
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_s
.Xauthority
Remote monitoring
---
apiVersion: v1
kind: Pod
metadata:
name: frontend
labels:
abc.com/name: nginx
abc.com/managed-by: manual
abc.com/version: "1"
abc.com/component: frontend
abc.com/part-of: webserver
abc.com/instance: august
spec:
containers:
- name: nginx
image: nginx
ports:
- containerPort: 80
Web server
Frontend
Backend


kubernetes recommends
Labels for Resources
1. networknuts.net/name=nginx
name
2. networknuts.net/managed-by=helm
managed by
3. networknuts.net/version="1.8.8"
version
4. networknuts.net/component=frontend
component
5. networknuts.net/part-of=webserver
part of
6. networknuts.net/instance=nginx-august
instance
prefix (optional) upto 253 chrs
key upto 63 chrs
value upto 63 chrs
#networknuts
192.168.121.152 (alok)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
2. 192.168.121.152 (alok)
/home/alok/
Name
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.Xauthority
Remote monitoring
Follow terminal folc
---
apiVersion: v1
kind: Pod
metadata:
name: frontend
labels:
abc.com/name: nginx
abc.com/managed-by: manual
abc.com/version: "1"
abc.com/component: frontend
abc.com/part-of: webserver
abc.com/instance: august
spec:
containers:
- name: nginx
image: nginx
ports:
- containerPort: 80
~
~
Web server
Frontend
manager 7% 1.19 GB / 3.79 GB 0.05 Mb/s 0.01 Mb/s 74 min alok (x3) /: 65% /boot: 14%


kubernetes recommends
Labels for Resources
1. networknuts.net/name=nginx
name
2. networknuts.net/managed-by=helm
managed by
3. networknuts.net/version="1.8.8"
version
4. networknuts.net/component=frontend
component
5. networknuts.net/part-of=webserver
part of
6. networknuts.net/instance=nginx-august
instance
prefix (optional) upto 253 chrs
key upto 63 chrs
value upto 63 chrs
#networknuts
192.168.121.152 (alok)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
2. 192.168.121.152 (alok)
/home/alok/
Name
.cache
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_s
.Xauthority
Remote monitoring
---
apiVersion: v1
kind: Pod
metadata:
name: backend
labels:
abc.com/name: redis
abc.com/managed-by: manual
abc.com/version: "2"
abc.com/component: backend
abc.com/part-of: webserver
abc.com/instance: july
spec:
containers:
- name: redis
image: redis
~
~
~
:wq!

```
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f yourpod.yaml
pod/frontend created
root@manager:~/weekends# kubectl create -f your-backend-pod.yaml
pod/backend created
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods --show-labels
NAME        READY   STATUS              RESTARTS   AGE   LABELS
backend     0/1     ContainerCreating   0          6s    abc.com/component=backend,abc.com/instance=july,abc.com/managed-by=manual,a
bc.com/name=redis,abc.com/part-of=webserver,abc.com/version=2
frontend    0/1     ContainerCreating   0          14s   abc.com/component=frontend,abc.com/instance=august,abc.com/managed-by=manua
l,abc.com/name=nginx,abc.com/part-of=webserver,abc.com/version=1
webserver   1/1     Running             0          14m   app=web,client=abc,env=prod
root@manager:~/weekends#
```

```
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f yourpod.yaml
pod/frontend created
root@manager:~/weekends# kubectl create -f your-backend-pod.yaml
pod/backend created
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods --show-labels
NAME        READY   STATUS              RESTARTS   AGE   LABELS
backend     0/1     ContainerCreating   0          6s    abc.com/component=backend,abc.com/instance=july,abc.com/managed-by=manual,a
bc.com/name=redis,abc.com/part-of=webserver,abc.com/version=2
frontend    0/1     ContainerCreating   0          14s   abc.com/component=frontend,abc.com/instance=august,abc.com/managed-by=manua
l,abc.com/name=nginx,abc.com/part-of=webserver,abc.com/version=1
webserver   1/1     Running             0          14m   app=web,client=abc,env=prod
root@manager:~/weekends#
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -l "abc.com/part-of=webserver"
NAME       READY   STATUS    RESTARTS   AGE
backend    1/1     Running   0          44s
frontend   1/1     Running   0          52s
root@manager:~/weekends#
root@manager:~/weekends# kubectl get pods -l "abc.com/part-of=webserver"  --show-labels
NAME       READY   STATUS    RESTARTS   AGE   LABELS
backend    1/1     Running   0          52s   abc.com/component=backend,abc.com/instance=july,abc.com/managed-by=manual,abc.com/name
=redis,abc.com/part-of=webserver,abc.com/version=2
frontend   1/1     Running   0          60s   abc.com/component=frontend,abc.com/instance=august,abc.com/managed-by=manual,abc.com/n
ame=nginx,abc.com/part-of=webserver,abc.com/version=1
root@manager:~/weekends#
```


01
kubectl get pods -l "app=apache" --show-labels
show the pods where label app=apache is found
02
kubectl get pods -l "app=nginx,env=prod" --show-labels
show the pods where label app=nginx "AND" env=prod. Comma is AND.
03
kubectl get pods -l "app=apache,client!=abc" --show-labels
show the pods where label app is apache "AND" client not abc
04
kubectl get pods -l "app in (apache,nginx)" --show-labels
show the pods where label app has either value apache OR nginx
#networknuts